



MODULE 3 | PART 3

TRIGONOMETRY

IOQM & RMO 2026

TRIGONOMETRY

Arcs and Angles

In a circle of radius r , an arc of length r will subtend an angle of 1 radian. An arc of length l will subtend an angle whose measure is $\frac{l}{r}$ radians.

Thus, if in a circle of radius r , an arc of length l subtends an angle θ radians at the centre, we have

$$\theta = \frac{l}{r} \text{ or } l = r\theta$$

Note:

$$\diamond 2\pi \text{ radians} = 360^\circ$$

$$\diamond \pi \text{ radians} = 180^\circ$$

$$1 \text{ radian} = \left(\frac{180}{\pi}\right)^\circ$$

$$1^\circ = \frac{\pi}{180} \text{ radians}$$



Conversion

Converting radians to degrees

$$\theta \text{ radians} = \left(\frac{180\theta}{\pi}\right)^\circ$$

Converting degrees to radians

$$\theta^\circ = \frac{\theta\pi}{180} \text{ radians}$$



1. If the arcs of same length in two circles subtend angles of 60° and 75° at their centres, find the ratio of their radii.





2. Three planets orbit a star circularly in the same plane. Each moves in the same direction and moves at constant speed. Their periods are 60, 84, and 140 years.

The three planets and the star are currently collinear. What is the fewest number of years from now that they will all be collinear again?

[AIME]

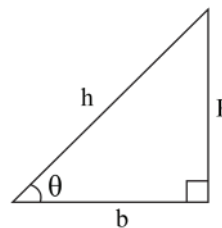


3. *An ant leaves the anthill for its morning exercise. It walks 4 feet east and then makes a 160° turn to the right and walks 4 more feet. It then makes another 160° turn to the right and walks 4 more feet. If the ant continues this pattern until it reaches the anthill again, what is the distance in feet it would have walked?

[PRMO 2019]



Trigonometric Ratios (or Trigonometric Functions)



In a right-angled triangle

$$\sin \theta = \frac{p}{h}; \cos \theta = \frac{b}{h}; \tan \theta = \frac{p}{b};$$

$$\operatorname{cosec} \theta = \frac{h}{p}; \sec \theta = \frac{h}{b} \text{ and } \cot \theta = \frac{b}{p}$$

where p is perpendicular, b is base, and h is hypotenuse.

Important Note Related to Regular Polygons

Let $A_1, A_2, \dots, A_n$ be vertices of a regular polygon inscribed in a circle of radius R .

$$\diamond A_1A_2 = 2R \sin \left(\frac{\pi}{n} \right)$$

$$\diamond A_1A_3 = 2R \sin \left(\frac{2\pi}{n} \right)$$

$$\diamond A_1A_4 = 2R \sin \left(\frac{3\pi}{n} \right), \dots$$

$$\diamond \text{Central angle } \theta = \frac{2\pi}{n}$$

$\diamond$ Diameter diagonal exist when n is even.





4. *Let $ABCD$ be a convex cyclic quadrilateral. Suppose P is a point in the plane of the quadrilateral such that the sum of its distances from the vertices of $ABCD$ is the least. If $\{PA, PB, PC, PD\} = \{3, 4, 6, 8\}$. What is the maximum possible area of $ABCD$?

[PRMO 2019]



5. If $\cos x + \cos^2 x = 1$ then value of $\sin^2 x + \sin^4 x$ will be divisible by

- (A) 1 (B) 2
(C) 4 (D) 3

Basic Trigonometric Identities

$$\diamond \sin \theta \cdot \operatorname{cosec} \theta = 1$$

$$\diamond \cos \theta \cdot \sec \theta = 1$$

$$\diamond \tan \theta \cdot \cot \theta = 1$$

$$\diamond \tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\diamond \cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\diamond \sin^2 \theta + \cos^2 \theta = 1 \text{ or}$$

$$\sin^2 \theta = 1 - \cos^2 \theta \text{ or}$$

$$\cos^2 \theta = 1 - \sin^2 \theta$$

$$\diamond \sec^2 \theta - \tan^2 \theta = 1 \text{ or}$$

$$\sec^2 \theta = 1 + \tan^2 \theta \text{ or}$$

$$\tan^2 \theta = \sec^2 \theta - 1$$



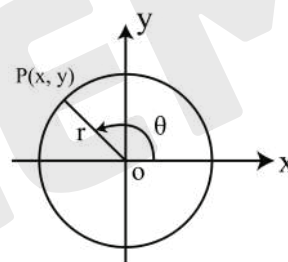
6. *Suppose that $\sec x + \tan x = \frac{22}{7}$ and that $\operatorname{cosec} x + \cot x = \frac{m}{n}$, where $\frac{m}{n}$ is in lowest terms. Find $m + n$.

[AIME]



Trigonometric Ratios Through Cartesian Coordinates

By using rectangular coordinates, the definitions of trigonometric functions can be extended to angles of any size in the following way (see diagram). A point P is taken with coordinates (x, y) . The radius vector OP has length r and the angle θ is taken as the directed angle measured anticlockwise from the x -axis. The three main trigonometric functions are then defined in terms of r and the coordinates x and y .



$$\sin \theta = \frac{y}{r}$$

$$\cos \theta = \frac{x}{r}$$

$$\tan \theta = \frac{y}{x},$$

(The other functions are reciprocals of these)

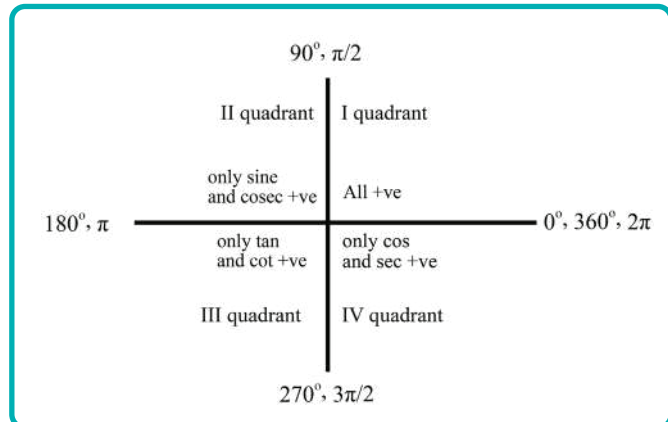
This can give negative values of the trigonometric functions.



7. *Let $\triangle ABC$ be a right triangle with right angle at C . Let D and E be points on $\overline{AB}$ with D between A and E such that $\overline{CD}$ and $\overline{CE}$ trisect $\angle C$. If $\frac{DE}{BE} = \frac{8}{15}$, then $\tan B$ can be written as $\frac{m\sqrt{p}}{n}$, where m and n are relatively prime positive integers, and p is a positive integer not divisible by the square of any prime. Find $m + n + p$.



Signs of Trigonometric Functions in Different Quadrants



Trigonometric Functions of Allied Angles

- ❖ $\sin(2n\pi + \theta) = \sin \theta, n \in \mathbb{Z}$
- ❖ $\cos(2n\pi + \theta) = \cos \theta, \text{ where } n \in \mathbb{Z}$
- ❖ $\sin(-\theta) = -\sin \theta$
- ❖ $\cos(-\theta) = \cos \theta$
- ❖ $\sin(90^\circ - \theta) = \cos \theta$
- ❖ $\cos(90^\circ - \theta) = \sin \theta$
- ❖ $\sin(90^\circ + \theta) = \cos \theta$
- ❖ $\cos(90^\circ + \theta) = -\sin \theta$
- ❖ $\sin(180^\circ - \theta) = \sin \theta$
- ❖ $\cos(180^\circ - \theta) = -\cos \theta$
- ❖ $\sin(180^\circ + \theta) = -\sin \theta$
- ❖ $\cos(180^\circ + \theta) = -\cos \theta$
- ❖ $\sin(270^\circ - \theta) = -\cos \theta$
- ❖ $\cos(270^\circ - \theta) = -\sin \theta$

- ❖ $\sin(270^\circ + \theta) = -\cos \theta$
- ❖ $\cos(270^\circ + \theta) = \sin \theta$
- ❖ $\sin(360^\circ - \theta) = -\sin \theta$
- ❖ $\cos(360^\circ - \theta) = \cos \theta$
- ❖ $\sin(360^\circ + \theta) = \sin \theta$
- ❖ $\cos(360^\circ + \theta) = \cos \theta$

Values of Trigonometric Ratio of Some Standard Angles

	0°	30°	45°	60°	90°	180°	270°
Angles T-ratio	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	π	$3\pi/2$
$\sin \theta$	0	1/2	$1/\sqrt{2}$	$\sqrt{3}/2$	1	0	-1
$\cos \theta$	1	$\sqrt{3}/2$	$1/\sqrt{2}$	1/2	0	-1	0
$\tan \theta$	0	$1/\sqrt{3}$	1	$\sqrt{3}$	N.D.	0	N.D.
$\cot \theta$	N.D.	$\sqrt{3}$	1	$1/\sqrt{3}$	0	N.D.	0
$\sec \theta$	1	$2/\sqrt{3}$	$\sqrt{2}$	2	N.D.	-1	N.D.
$\csc \theta$	N.D.	2	$\sqrt{2}$	$2/\sqrt{3}$	1	N.D.	-1

N.D. $\rightarrow$ Not Defined

- ❖ $\sin n\pi = 0;$
- $\cos n\pi = (-1)^n;$
- $\tan n\pi = 0, \text{ where } n \in \mathbb{Z}$
- ❖ $\sin(2n+1)\frac{\pi}{2} = (-1)^n;$
- $\cos(2n+1)\frac{\pi}{2} = 0, \text{ where } n \in \mathbb{Z}$



8. If $x \sin 45^\circ \cos^2 60^\circ = \frac{\tan^2 60^\circ \operatorname{cosec} 30^\circ}{\sec 45^\circ \cot^2 30^\circ}$,
then $x =$

- (A) 2 (B) 4
(C) 8 (D) 16

[KEAM 2002]



9. The value of $\cos 1^\circ \cos 2^\circ \cos 3^\circ \cdots \cos 179^\circ$
is equal to

- (A) $\frac{1}{\sqrt{2}}$ (B) 0
(C) 1 (D) -1

[KEAM 2001, NCERT EP]



Trigonometric Ratios of the Sum and Difference of Two Angles

$$\diamond \sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\diamond \sin(A - B) = \sin A \cos B - \cos A \sin B$$

$$\diamond \cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\diamond \cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$\diamond \tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\diamond \tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$\diamond \cot(A + B) = \frac{\cot B \cot A - 1}{\cot B + \cot A}$$

$$\diamond \cot(A - B) = \frac{\cot B \cot A + 1}{\cot B - \cot A}$$



10. If $\sin \alpha + \sin \beta = \frac{\sqrt{6}}{2}$ and

$\cos \alpha + \cos \beta = \frac{\sqrt{2}}{2}$, then $\cos(\alpha - \beta)$ is equal to

- (A) $\frac{1}{2}$ (B) $\frac{3}{2}$
(C) $-\frac{1}{2}$ (D) 0

[KEAM 2021]



11. Prove that $\sqrt{3} \operatorname{cosec} 20^\circ - \sec 20^\circ = 4$.



12. $\frac{\tan 2\alpha + \tan 3\alpha - \tan 5\alpha}{\tan 2\alpha \cdot \tan 3\alpha \cdot \tan 5\alpha}$ is equal to
 (A) $\tan \alpha$ (B) $\cot \alpha$
 (C) -1 (D) 1



13. In triangle ABC , $\tan \angle CAB = \frac{22}{7}$, and the altitude from A divides BC into segments of length 3 and 17. What is the area of triangle ABC ?

[AIME]



Transformation Formulae

- ❖ $\cos(A + B) + \cos(A - B) = 2 \cos A \cos B$
- ❖ $\cos(A + B) - \cos(A - B) = -2 \sin A \sin B$
- ❖ $\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$
- ❖ $\sin(A + B) - \sin(A - B) = 2 \cos A \sin B$
- ❖ $\cos C + \cos D$
 $= 2 \cos \left(\frac{C + D}{2} \right) \cos \left(\frac{C - D}{2} \right)$
- ❖ $\cos C - \cos D$
 $= -2 \sin \left(\frac{C + D}{2} \right) \sin \left(\frac{C - D}{2} \right)$
- ❖ $\sin C + \sin D$
 $= 2 \sin \left(\frac{C + D}{2} \right) \cos \left(\frac{C - D}{2} \right)$
- ❖ $\sin C - \sin D$
 $= 2 \sin \left(\frac{C - D}{2} \right) \cos \left(\frac{C + D}{2} \right)$



14. $\frac{\cos 7x + \cos 5x}{\sin 7x - \sin 5x} =$

- (A) $\tan x$ (B) $\cot x$
(C) $\tan 2x$ (D) $\cot 2x$

[NCERT TB]



Some more results:

$$\begin{aligned} \diamond \sin^2 A - \sin^2 B &= \sin(A+B) \cdot \sin(A-B) \\ &= \cos^2 B - \cos^2 A. \end{aligned}$$

$$\begin{aligned} \diamond \cos^2 A - \sin^2 B &= \cos(A+B) \cdot \cos(A-B). \end{aligned}$$

Maximum and Minimum Values of $a \sin x + b \cos x + c$

$$c - \sqrt{a^2 + b^2} \leq a \sin x + b \cos x + c \leq c + \sqrt{a^2 + b^2}$$



15. The minimum value of $3 \cos x + 4 \sin x + 8$ is

- (A) 5 (B) 9
(C) 7 (D) 3

[NCERT EP]



16. *Given that $(1 + \sin t)(1 + \cos t) = \frac{5}{4}$ and $(1 - \sin t)(1 - \cos t) = \frac{m}{n} - \sqrt{k}$, where k, m and n are positive integers with m and n relatively prime, find $k + m + n$.

[AIME]



Double Angle Formulae

$$\diamond \sin 2\theta = 2 \sin \theta \cos \theta = \frac{2 \tan \theta}{1 + \tan^2 \theta}$$

$$\begin{aligned} \diamond \cos 2\theta &= \cos^2 \theta - \sin^2 \theta \\ &= 2 \cos^2 \theta - 1 \\ &= 1 - 2 \sin^2 \theta \\ &= \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \end{aligned}$$

$$\diamond 1 + \cos 2\theta = 2 \cos^2 \theta$$

$$\diamond 1 - \cos 2\theta = 2 \sin^2 \theta$$

$$\diamond \frac{1 - \cos 2\theta}{1 + \cos 2\theta} = \tan^2 \theta$$

$$\diamond \tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

Half Angle Formulae

$$\diamond \sin \theta = 2 \sin \left(\frac{\theta}{2} \right) \cos \left(\frac{\theta}{2} \right)$$

$$= \frac{2 \tan \left(\frac{\theta}{2} \right)}{1 + \tan^2 \left(\frac{\theta}{2} \right)}$$

$$\diamond \tan \theta = \frac{2 \tan \left(\frac{\theta}{2} \right)}{1 - \tan^2 \left(\frac{\theta}{2} \right)}$$

$$\diamond \sin^2 \left(\frac{\theta}{2} \right) = \frac{1 - \cos \theta}{2}$$

$$\diamond \cos^2 \left(\frac{\theta}{2} \right) = \frac{1 + \cos \theta}{2}$$

$$\diamond \frac{1 + \cos \theta}{\sin \theta} = \frac{\sin \theta}{1 - \cos \theta} = \cot \left(\frac{\theta}{2} \right)$$

$$\diamond \frac{1 - \cos \theta}{\sin \theta} = \frac{\sin \theta}{1 + \cos \theta} = \tan \left(\frac{\theta}{2} \right)$$



17. *If $x = \cos 1^\circ \cos 2^\circ \cos 3^\circ \dots \cos 89^\circ$ and $y = \cos 2^\circ \cos 6^\circ \cos 10^\circ \dots \cos 86^\circ$, then what is the integer nearest to $\frac{2}{7} \log_2 \left(\frac{y}{x} \right)$?

[PRMO 2018]



Triple Angle Formulae

$$\diamond \sin 3x = 3 \sin x - 4 \sin^3 x$$

$$\diamond \cos 3x = 4 \cos^3 x - 3 \cos x$$

$$\diamond \tan 3x = \frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x}$$



18. The value of $\frac{\cos^3 10^\circ + \sin^3 20^\circ}{\cos 10^\circ + \sin 20^\circ}$ is equal to

(A) $\frac{1}{2}$

(B) $\frac{3}{4}$

(C) $\frac{2}{3}$

(D) $\frac{1}{4}$



Trigonometric Functions of Some Common Angles

$$\diamond \tan 22.5^\circ = \sqrt{2} - 1$$

$$\diamond \tan 15^\circ = 2 - \sqrt{3}$$

$$\diamond \cot 15^\circ = 2 + \sqrt{3}$$

$$\diamond \sin 15^\circ = \frac{\sqrt{3} - 1}{2\sqrt{2}}$$

$$\diamond \cos 15^\circ = \frac{\sqrt{3} + 1}{2\sqrt{2}}$$

$$\diamond \sin 18^\circ = \frac{\sqrt{5} - 1}{4}$$

$$\diamond \cos 36^\circ = \frac{\sqrt{5} + 1}{4}$$



19. $\sin 47^\circ + \sin 61^\circ - \sin 11^\circ - \sin 25^\circ$ is equal to

- (A) $\sin 36^\circ$ (B) $\cos 36^\circ$
(C) $\sin 7^\circ$ (D) $\cos 7^\circ$



Note:

$$\diamond \sin A \sin(60 + A) \sin(60 - A) = \frac{1}{4} \sin 3A$$

$$\diamond \cos A \cos(60 + A) \cos(60 - A) = \frac{1}{4} \cos 3A$$

$$\diamond \tan A \tan(60 + A) \tan(60 - A) = \tan 3A$$

Example:

If $\sin 6^\circ \sin 42^\circ \sin 54^\circ \sin 66^\circ \sin 78^\circ = \frac{\sqrt{a} + b}{c}$, where a, b and c are relatively prime numbers, then value of $(a + b + c)$ is:

- (A) 10 (B) 22
(C) 20 (D) 70

Solution:

$$\begin{aligned} & (\sin 6^\circ \sin 54^\circ \sin 66^\circ) \sin 42^\circ \sin 78^\circ \\ &= \frac{1}{4} (\sin 18^\circ \sin 42^\circ \sin 78^\circ) \\ &= \frac{1}{16} \sin 54^\circ = \frac{\sqrt{5} + 1}{64} \end{aligned}$$



20. $16 \sin (20^\circ) \sin (40^\circ) \sin (80^\circ)$ is equal to

- (A) $\sqrt{3}$ (B) $2\sqrt{3}$
(C) 3 (D) $4\sqrt{3}$

[JEE Main 2022]



Sum of Sines or Cosines of an AP

$$\diamond \sin a + \sin(a + d) + \sin(a + 2d) + \dots$$

$$\dots + \sin(a + (n - 1)d)$$

$$= \frac{\sin\left(a + (n - 1)\frac{d}{2}\right) \sin\left(\frac{nd}{2}\right)}{\sin\left(\frac{d}{2}\right)}$$

$$\diamond \cos a + \cos(a + d) + \cos(a + 2d) + \dots$$

$$\dots + \cos(a + (n - 1)d)$$

$$= \frac{\cos\left(a + (n - 1)\frac{d}{2}\right) \sin\left(\frac{nd}{2}\right)}{\sin\left(\frac{d}{2}\right)}$$



21. Let $x = \frac{\sum_{n=1}^{44} \cos n^\circ}{\sum_{n=1}^{44} \sin n^\circ}$. What is the greatest integer that does not exceed $100x$?

[AIME]



22. Given that $\sum_{k=1}^{35} \sin 5k = \tan \frac{m}{n}$, where angles are measured in degrees, and m and n are relatively prime positive integers that satisfy $\frac{m}{n} < 90^\circ$ find $m + n$.

[AIME]



Telescoping Series Problems



23. The value of

$$\sum_{k=1}^{13} \frac{1}{\sin\left(\frac{\pi}{4} + \frac{(k-1)\pi}{6}\right) \sin\left(\frac{\pi}{4} + \frac{k\pi}{6}\right)}$$

is equal to

- (A) $3 - \sqrt{3}$ (B) $2(3 - \sqrt{3})$
(C) $2(\sqrt{3} - 1)$ (D) $2(2 - \sqrt{3})$

[JEE Adv 2016]



24. *Find the least positive integer n such that

$$\frac{1}{\sin 45^\circ \sin 46^\circ} + \frac{1}{\sin 47^\circ \sin 48^\circ} + \dots + \frac{1}{\sin 133^\circ \sin 134^\circ} = \frac{1}{\sin n^\circ}$$

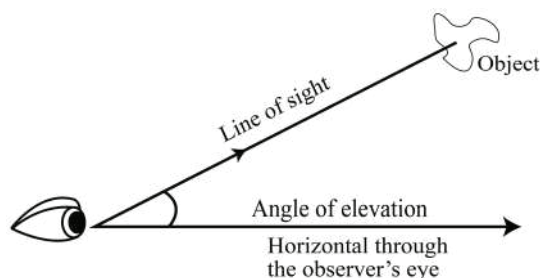
[AIME]



Angle of Elevation

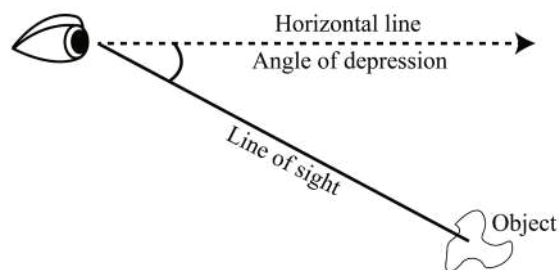
In order to see an object that is at a higher level compared to the ground level, we have to look up. The line joining the object and the eye of the observer is known as the line of sight and the angle that this line of sight makes with the horizontal drawn through the eye of the observer is known as the **angle of elevation**.

Therefore, the angle of elevation of an object helps in finding out its height (Figure).



Angle of Depression

When the object is at a lower level than the observer's eyes, he has to look downwards to have a view of the object. In that case, the angle that the line of sight makes with the horizontal through the observer's eye is known as the **angle of depression** (Figure).



25. A man is standing on the deck of a ship, which is 8 m above water level. He observes the angle of elevation of the top of a hill as 60° and the angle of depression of the base of the hill as 30° . Calculate the distance of the hill from the ship and the height of the hill.



Inverse Trigonometric Functions

Functions	Domain	Range (Principal value branches)
$y = \sin^{-1} x$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$
$y = \cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$y = \operatorname{cosec}^{-1} x$	$\mathbb{R} - (-1, 1)$	$[-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$
$y = \sec^{-1} x$	$\mathbb{R} - (-1, 1)$	$[0, \pi] - \{\frac{\pi}{2}\}$
$y = \tan^{-1} x$	$\mathbb{R}$	$(-\frac{\pi}{2}, \frac{\pi}{2})$
$y = \cot^{-1} x$	$\mathbb{R}$	$(0, \pi)$

❖ $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$

❖ $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$

❖ $\sec^{-1} x + \operatorname{cosec}^{-1} x = \frac{\pi}{2}$



26. The principal value of $\cos^{-1} x$, for $x = \frac{\sqrt{3}}{2}$ is

- (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$
 (C) $\frac{2\pi}{3}$ (D) $\frac{5\pi}{6}$

[NCERT EP]



27. The value of $\sin^{-1} \left[\cos \left(\sin^{-1} \frac{\sqrt{3}}{2} \right) \right]$ is

- (A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{2}$
 (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{3}$

[NCERT EP]



28. *Let AB and CD be two parallel chords in a circle with radius 5 such that the centre O lies between these chords. Suppose $AB = 6$, $CD = 8$. Suppose further that the area of the part of the circle lying between the chords AB and CD is $\left(\frac{m\pi + n}{k} \right)$, where m, n, k are positive integers with $\gcd(m, n, k) = 1$. What is the value of $m + n + k$?

[PRMO 2017]



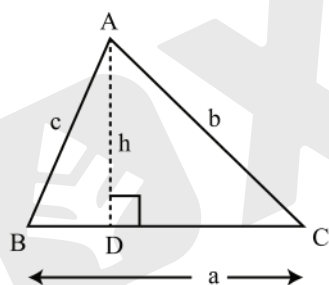
Solution of Triangle

Sine Formulae

In any triangle ABC

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = \lambda = \frac{abc}{2\Delta} = 2R$$

where R is circumradius and Δ is area of the triangle.



29. In triangle ABC , angles A and B measure 60 degrees and 45 degrees, respectively. The bisector of angle A intersects $\overline{BC}$ at T , and $AT = 24$. The area of triangle ABC can be written in the form $a + b\sqrt{c}$, where a, b , and c are positive integers, and c is not divisible by the square of any prime. Find $a + b + c$.

[AIME]



30. *In a triangle ABC , it is known that $\angle A = 100^\circ$ and $AB = AC$. The internal angle bisector BD has length 20 units. Find the length of BC to the nearest integer, given that $\sin 10^\circ \approx 0.174$.

[PRMO 2019]



Cosine Formulae

$$\diamond \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\diamond \cos B = \frac{c^2 + a^2 - b^2}{2ca}$$

$$\diamond \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\diamond a^2 = b^2 + c^2 - 2bc \cos A$$



31. Let a, b, c be the three sides of a triangle, and let α, β, γ be the angles opposite them. If $a^2 + b^2 = 1989c^2$, find $\frac{\cot \gamma}{\cot \alpha + \cot \beta}$.

[AIME]



32. *In equilateral $\triangle ABC$ let points D and E trisect $\overline{BC}$. Then $\sin(\angle DAE)$ can be expressed in the form $\frac{a\sqrt{b}}{c}$, where a and c are relatively prime positive integers, and b is an integer that is not divisible by the square of any prime. Find $a + b + c$.

[AIME]



Projection Formulae

$$a = b \cos C + c \cos B$$

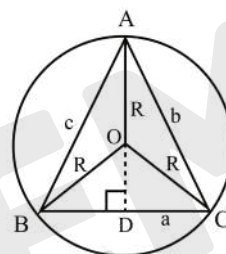
Half Angle Formulae

$$\sin \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}$$

$$\cos \frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}$$

$$\tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$$

Radius of the Circumcircle R

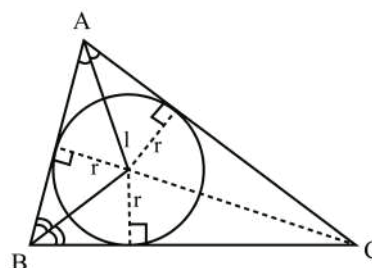


Circumcentre is the point of intersection of perpendicular bisectors of the sides and distance between circumcentre and vertex of triangle is called circumradius ' R '.

$$R = \frac{a}{2 \sin A} = \frac{b}{2 \sin B} = \frac{c}{2 \sin C} = \frac{abc}{4\Delta}$$

Radius of the Incircle r

Point of intersection of internal angle bisectors is incentre and perpendicular distance of incentre from any side is called inradius r .



$$r = \frac{\Delta}{s}$$



33. In a triangle ABC , if $a : b : c = 4 : 5 : 6$, then ratio between its circumradius and inradius is:

- (A) $\frac{16}{7}$ (B) $\frac{16}{9}$
(C) $\frac{7}{16}$ (D) $\frac{11}{7}$



34. *In a circle, parallel chords of lengths 2, 3 and 4 determine central angles of α, β and $\alpha + \beta$ radians, respectively, where $\alpha + \beta < \pi$. If $\cos \alpha$, which is a positive rational number, is expressed as a fraction in lowest terms, what is the sum of its numerator and denominator?

[AIME]



Area of triangle

❖ Heron's Formula

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

where $s =$ semi perimeter

❖ $\Delta = \frac{1}{2}ab \sin C$

❖ $\Delta = r \cdot s$, where $r =$ inradius.

❖ $\Delta = \frac{abc}{4R}$, where $R =$ circumradius



35. *The sides x and y of a scalene triangle satisfy $x + \frac{2\Delta}{x} = y + \frac{2\Delta}{y}$, where Δ is the area of the triangle. If $x = 60, y = 63$, what is the length of the largest side of the triangle?

[IOQM 2021]



ANSWER KEY

(1) 5 : 4	(8) (C)	(15) (D)	(22) 177	(29) 291
(2) 105	(9) (B)	(16) 27	(23) (C)	(30) 27
(3) 36	(10) (D)	(17) 19	(24) 1	(31) 994
(4) 55	(11)	(18) (B)	(25) $(8\sqrt{3}, 32)$	(32) 20
(5) (A)	(12) (C)	(19) (D)	(26) (A)	(33) (A)
(6) 44	(13) 110	(20) (B)	(27) (C)	(34) 49
(7) 18	(14) (B)	(21) 241	(28) 75	(35) 87

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